

# SlabMaster™ Enterprise Platform

## Technical Specification Document (TSD)

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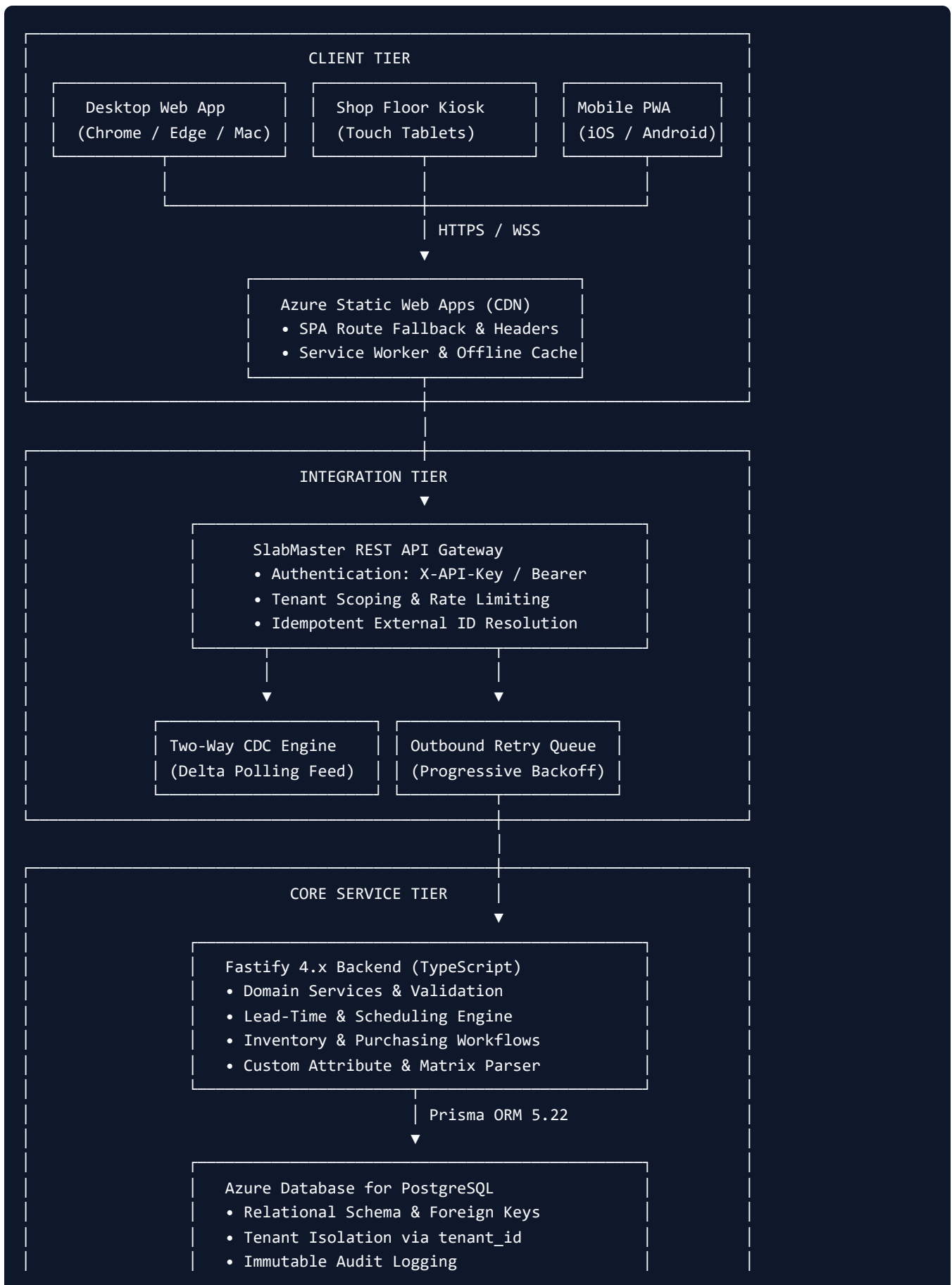
**Classification:** Enterprise System Architecture & Engineering Specification

**Status:** Implemented & Production-Ready

**Updated:** September 2026

### 1. System Topology & Architecture

SlabMaster is architected as a high-throughput, multi-tenant cloud application combining a lightweight, offline-first React Progressive Web App (PWA) on the client tier with an event-driven, high-performance Node.js / Fastify microservice on the backend tier.



## 2. Technology Stack & Component Specifications

| Tier               | Component          | Technology Selection                         | Justification / Purpose                                                                 |
|--------------------|--------------------|----------------------------------------------|-----------------------------------------------------------------------------------------|
| Frontend Framework | Client UI          | React 18.3+ & TypeScript 5.5+                | Declarative component hierarchy, strict type safety, fast reconciliation.               |
| Build & Bundler    | Build Tooling      | Vite 5.4+                                    | Sub-second HMR, optimized Rollup production builds, dynamic code splitting.             |
| Styling & Theme    | UI Presentation    | Tailwind CSS 3.4+                            | Utility-first styling with native high-contrast Dark Mode and mobile responsiveness.    |
| Icons & Media      | Iconography        | Lucide React 0.344+                          | Tree-shakable SVG icon system matching enterprise design standards.                     |
| Offline Storage    | PWA Client DB      | IndexedDB / Dexie.js & Service Worker        | Local persistence of work orders, customer signatures, and photos during field outages. |
| Backend Framework  | Application Server | Fastify 4.28+ (Node.js 20 LTS)               | Low-overhead, high-throughput (30k+ req/sec), JSON schema-driven routing.               |
| ORM & Migrations   | Data Access        | Prisma ORM 5.22+                             | Type-safe query builder, declarative schema migrations, automated TypeScript client.    |
| Database           | Persistence Store  | PostgreSQL 15+ / Azure Database              | ACID compliance, JSONB support for dynamic attributes, relational foreign keys.         |
| Authentication     | Enterprise SSO     | Microsoft Entra ID (Azure AD MSAL)           | Corporate single sign-on, JWT token verification, RBAC claims.                          |
| Hosting & CDN      | Infrastructure     | Azure Static Web Apps + Azure Container Apps | Global edge distribution, automated Git CI/CD triggers, zero-downtime deployments.      |
| Testing            | Automated QA       | Vitest 2.1+                                  | Native Vite test runner executing in CI/CD pipeline during Azure builds.                |

## 3. Database Schema Architecture (Prisma ORM)

### 3.1 Relational Entity Model Overview

```
erDiagram
    SubscriberTenant ||--o{ User : "has many"
    SubscriberTenant ||--o{ Account : "owns"
    SubscriberTenant ||--o{ ApiKey : "provisions"
    SubscriberTenant ||--o{ ErpOutboundQueue : "queues"
    Account ||--o{ Community : "contains"
    Community ||--o{ Lot : "contains"
    Lot ||--o{ Job : "hosts"
    Job ||--o{ JobActivity : "sequences"
    Job ||--o{ Slab : "reserves"
    SubscriberTenant ||--o{ Slab : "inventories"
    SubscriberTenant ||--o{ PurchaseOrder : "issues"
    PurchaseOrder ||--o{ Slab : "receives"
```

### 3.2 Key Models & DDL Definitions

#### SubscriberTenant & ApiKey

```
model SubscriberTenant {
  id          String          @id @default(uuid())
  tenantName  String          @map("tenant_name")
  planTier    String          @default("ENTERPRISE") @map("plan_tier")
  createdAt   DateTime         @default(now()) @map("created_at")
  apiKeys     ApiKey[]
  erpQueue    ErpOutboundQueue[]
  accounts    Account[]
  slabs       Slab[]

  @@map("subscriber_tenants")
}

model ApiKey {
  id          String          @id @default(uuid())
  tenantId    String          @map("tenant_id")
  name        String
  keyHash     String          @map("key_hash")
  prefix      String          // e.g. "sm_live_9f83"
  scopes      String          @default("READ,WRITE,ERP_SYNC")
  lastUsedAt  DateTime?       @map("last_used_at")
  revokedAt   DateTime?       @map("revoked_at")
  createdAt   DateTime         @default(now()) @map("created_at")
  tenant      SubscriberTenant @relation(fields: [tenantId], references: [id], onDelete: Cascade)

  @@map("api_keys")
}
```

## ErpOutboundQueue (Progressive Backoff Retry Queue)

```
enum ErpSyncStatus {
  PENDING
  RETRYING
  FAILED
  COMPLETED
}

model ErpOutboundQueue {
  id          String          @id @default(uuid())
  tenantId    String          @map("tenant_id")
  entityType  String          @map("entity_type") // JOB, LOT, ACTIVITY, ACCOUNT
  entityId    String          @map("entity_id")
  action      String          // UPSERT, STATUS_UPDATE, CANCEL
  payload     Json
  targetUrl   String          @map("target_url")
  attempts    Int             @default(0)
  maxAttempts Int             @default(5) @map("max_attempts")
  status      ErpSyncStatus   @default(PENDING)
  lastError   String?         @map("last_error")
  nextRetryAt DateTime?       @map("next_retry_at")
  createdAt   DateTime        @default(now()) @map("created_at")
  updatedAt   DateTime        @updatedAt @map("updated_at")
  tenant      SubscriberTenant @relation(fields: [tenantId], references: [id], onDelete: Cascade)

  @@index([tenantId, status])
  @@index([nextRetryAt])
  @@map("erp_outbound_queue")
}
```

**3-Tier Hierarchy:** Account , Community , Lot , Job , JobActivity

```

model Account {
  id          String      @id @default(uuid())
  tenantId    String      @map("tenant_id")
  externalId  String?     @map("external_id")
  name        String
  accountType String      @default("Builder") @map("account_type")
  communities Community[]
  jobs        Job[]

  @@unique([tenantId, externalId])
  @@map("accounts")
}

model Community {
  id          String      @id @default(uuid())
  accountId   String      @map("account_id")
  externalId  String?     @map("external_id")
  name        String
  lots        Lot[]
  account     Account     @relation(fields: [accountId], references: [id], onDelete: Cascade)

  @@map("communities")
}

model Lot {
  id          String      @id @default(uuid())
  communityId String      @map("community_id")
  externalId  String?     @map("external_id")
  lotNumber   String      @map("lot_number")
  streetAddress String?   @map("street_address")
  jobs        Job[]
  community   Community   @relation(fields: [communityId], references: [id], onDelete: Cascade)

  @@map("lots")
}

model Job {
  id          String      @id @default(uuid())
  lotId       String      @map("lot_id")
  accountId   String      @map("account_id")
  externalId  String?     @map("external_id")
  jobName     String      @map("job_name")
  status      String      @default("Active")
  activities  JobActivity[]
  slabs       Slab[]
  lot         Lot          @relation(fields: [lotId], references: [id], onDelete: Cascade)
  account     Account      @relation(fields: [accountId], references: [id], onDelete: Restrict)

  @@unique([accountId, externalId])
  @@map("jobs")
}

```

```
model JobActivity {
  id          String  @id @default(uuid())
  jobId       String  @map("job_id")
  externalId  String? @map("external_id")
  activityType String  @map("activity_type")
  status      String  @default("Scheduled")
  scheduledDate DateTime? @map("scheduled_date")
  job         Job      @relation(fields: [jobId], references: [id], onDelete: Cascade)

  @@map("job_activities")
}
```

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## 4. REST API Gateway & Routing Specifications

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### 4.1 Authentication & Security Protocol

Requests require authentication using either:

- HTTP Header: `X-API-Key: sm_live_<token>`
- HTTP Header: `Authorization: Bearer <jwt_or_token>`

API keys are verified using SHA-256 hash comparison against `api_keys.key_hash`. The request context is decorated with `request.tenantId` to enforce strict multi-tenant isolation.

## 4.2 Endpoint Matrix

| Method | Endpoint Path                          | Idempotent | Description                                                                       |
|--------|----------------------------------------|------------|-----------------------------------------------------------------------------------|
| POST   | /api/v1/accounts/upsert                | Yes        | Upserts Builder Account by <code>external_id</code> .                             |
| GET    | /api/v1/accounts/by-external-id/:id    | Read       | Retrieves Account by ERP identifier.                                              |
| DELETE | /api/v1/accounts/by-external-id/:id    | Yes        | Soft-deactivates Account.                                                         |
| POST   | /api/v1/communities/upsert             | Yes        | Upserts Community linked to parent Account.                                       |
| GET    | /api/v1/communities/by-external-id/:id | Read       | Retrieves Community by ERP identifier.                                            |
| POST   | /api/v1/lots/upsert                    | Yes        | Upserts Lot linked to Community and Account.                                      |
| GET    | /api/v1/lots/by-external-id/:id        | Read       | Retrieves Lot details and active Jobs.                                            |
| POST   | /api/v1/jobs/upsert                    | Yes        | Upserts Job linked to Lot, Account, and Community.                                |
| GET    | /api/v1/jobs/by-external-id/:id        | Read       | Retrieves Job with activities and reserved slabs.                                 |
| DELETE | /api/v1/jobs/by-external-id/:id        | Yes        | Soft-cancels Job and releases reserved slabs.                                     |
| POST   | /api/v1/activities/upsert              | Yes        | Upserts milestone activity (Template, Sawjet, Install).                           |
| GET    | /api/v1/activities/by-external-id/:id  | Read       | Retrieves Activity status and scheduling.                                         |
| GET    | /api/v1/sync/changes                   | Read       | Polls Two-Way CDC event delta stream ( <code>since</code> , <code>limit</code> ). |
| GET    | /api/v1/sync/queue                     | Read       | Queries outbound ERP retry queue with status filters.                             |
| POST   | /api/v1/sync/queue/:id/retry           | Yes        | Resets a stuck dead-letter item to <code>PENDING</code> for immediate retry.      |
| POST   | /api/v1/sync/queue/retry-all           | Yes        | Resets all <code>FAILED</code> dead-letter queue items.                           |



## 5. Outbound Progressive Backoff & Dead-Letter Engine

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### 5.1 Mathematical Retry Calculation

When an outbound HTTP push to an external ERP endpoint fails (network timeout, 5xx server error, 429 rate limit), the queue service invokes `calculateNextRetryDelayMs(attempts)` :

$$\Delta t(\text{attempt}) = \begin{cases} 0 \text{ ms} & \text{if } \text{attempt} = 0 \text{ (Immediate retry)} \\ 60,000 \text{ ms} (1 \text{ min}) & \text{if } \text{attempt} = 1 \\ 300,000 \text{ ms} (5 \text{ min}) & \text{if } \text{attempt} = 2 \\ 900,000 \text{ ms} (15 \text{ min}) & \text{if } \text{attempt} = 3 \\ 3,600,000 \text{ ms} (1 \text{ hr}) & \text{if } \text{attempt} = 4 \\ \text{TERMINATE (Dead-Letter)} & \text{if } \text{attempt} \geq 5 \end{cases}$$

### 5.2 State Machine

- **Enqueue:** New events enter state `PENDING` with `attempts = 0`.
- **Failure < Max (5):** State transitions to `RETRYING`, `attempts += 1`, and `nextRetryAt = now() + \Delta t`.
- **Failure >= Max (5):** State transitions to `FAILED` (Dead-Letter), `nextRetryAt = null`, and diagnostic logs recorded in `lastError`.
- **Manual Trigger:** Admin clicking "Retry Now" transitions state back to `PENDING` with `nextRetryAt = now()`.

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## 6. Testing Architecture & Build Verification

The platform maintains a mandatory **100% test pass rate** enforced at build time via Vitest.

### Test Suite Inventory ( `frontend/src/tests/` ):

1. `erpQueueManagement.test.ts` (6 tests): Progressive backoff math, dead-letter state assignment, manual reset, Postman collection structure, API docs pack validation.
2. `apiKeysManagement.test.ts` (5 tests): API key generation, prefix masking, scope serialization, revocation logic.
3. `slabInventory.test.ts` (5 tests): Slab lifecycle transitions, remnant calculation, barcode formatting, bin allocation.
4. `purchasing.test.ts` (4 tests): PO creation, line item summation, receiving status transitions.
5. `formsTableMatrixRunner.test.ts` (4 tests): Table matrix row dynamic addition, column calculation, signature capture.
6. `customAttributesUtils.test.ts` (4 tests): Dynamic EAV attribute extraction, type validation, fallback values.
7. `tableMatrixUtils.test.ts` (3 tests): Table matrix serialization, column total formulas.
8. `shopFloor.test.ts` (4 tests): Station work queue sequencing, piece completion, one-tap milestone advance.

9. `helpDocAnchors.test.ts` (4 tests): Anchor link validation, help center table of contents parity.

**Total Automated Coverage:** 39 tests across 9 test suites passing with 0 failures.